FURTHER MECHANICS / KINEMATICS & DYNAMICS

Motion Under a Variable Force

Questions by topic 14 questions · 162 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*9 marks*	2
Question 2	*7 marks*	3
Question 3	*9 marks*	4
Question 4	*13 marks*	5
Question 5	*14 marks*	6
Question 6	*16 marks*	7
Question 7	*9 marks*	8
Question 8	*11 marks*	9
Question 9	*12 marks*	10
Question 10	*10 marks*	11
Question 11	*13 marks*	12
Question 12	*9 marks*	13
Question 13	*14 marks*	14
Question 14	*16 marks*	15

1. A particle P of mass $4\,\text{kg}$ moves along a straight horizontal line. At time $t\,\text{s}$, its displacement in the positive direction from a fixed point O is $x\,\text{m}$ and its velocity is $v\,\text{m}\,\text{s}^{-1}$.

The only horizontal force acting on P has magnitude

$$\frac{36}{(x+2)^3}\,\text{N}$$

and acts in the direction of decreasing x. Initially, P is at O and is moving in the positive direction with speed $\frac{3}{2}\,\text{m}\,\text{s}^{-1}$.

(a) Find an expression for v in terms of x. **[5]**

(b) Hence find an expression for x in terms of t. **[4]**

2. A lorry of mass 2000 kg travels along a straight horizontal road. Its engine delivers constant power of 80 kW. When the lorry is moving at speed $v\,\mathrm{m\,s^{-1}}$, the resistance to its motion has magnitude $200v\,\mathrm{N}$.

At time t seconds, the speed of the lorry is $v\,\mathrm{m\,s^{-1}}$. At $t = 0$, the speed of the lorry is $8\,\mathrm{m\,s^{-1}}$.

(a) Show that

$$10v\frac{\mathrm{d}v}{\mathrm{d}t} = 400 - v^2$$ **[3]**

(b) Verify that, while $8 \leq v < 20$,

$$t = 5\ln\left(\frac{336}{400 - v^2}\right)$$ **[4]**

3. A particle P of mass 0.5 kg moves along the positive x-axis under the action of a single force. At time t seconds, the particle is x metres from the origin O, where $x > 0$, and has speed $v\,\mathrm{m\,s^{-1}}$. The force acts in the direction of increasing x and has magnitude

$$\frac{k}{x^3} \text{ newtons}$$

where k is a positive constant.

When $x = 2$, the speed of P is $1\,\mathrm{m\,s^{-1}}$, and when $x = 4$, its speed is $2\,\mathrm{m\,s^{-1}}$.

(a) Show that

$$v^2 = 5 - \frac{16}{x^2}$$ **[5]**

The time taken for the speed of P to increase from $1\,\mathrm{m\,s^{-1}}$ to $2\,\mathrm{m\,s^{-1}}$ is T seconds.

(b) Use algebraic integration to show that

$$T = \frac{6}{5}$$ **[4]**

4. A particle P of mass $3\,\text{kg}$ moves along the positive x-axis. At time $t = 0$, it passes through the origin with speed $4\,\text{m}\,\text{s}^{-1}$.

While P is moving, the only force acting on it is a resistive force of magnitude

$$\frac{6}{v}\,\text{N}$$

where $v\,\text{m}\,\text{s}^{-1}$ is its speed. The force acts in the negative x-direction.

(a) Find an expression for v in terms of the displacement x of P from the origin. **[5]**

(b) Find the distance travelled by P while its speed decreases from $4\,\text{m}\,\text{s}^{-1}$ to $2\,\text{m}\,\text{s}^{-1}$. **[2]**

A particle Q, also of mass $3\,\text{kg}$, starts from the origin with speed $4\,\text{m}\,\text{s}^{-1}$. The only force acting on Q is a constant resistive force of magnitude $\frac{3}{2}\,\text{N}$.

(c) By comparing the resistive forces acting on P and Q, explain why P must come to rest at a time $t < 8\,\text{s}$ and at a displacement $x < 16\,\text{m}$. **[3]**

(d) Sketch the velocity–time graph for P from $t = 0$ until it first comes to rest. Your sketch should indicate the concavity of the graph, but no numerical values are required. **[1]**

(e) Determine the maximum displacement of P from the origin. **[2]**

5. A particle B of mass $2\,\text{kg}$ moves along the positive x-axis. When B passes through the origin, its speed is $u\,\text{m s}^{-1}$.

The only horizontal force acting on B is a resistance whose magnitude is directly proportional to the cube of the speed of B. When the speed of B is $3\,\text{m s}^{-1}$, the magnitude of the resistance is $54\,\text{N}$.

(a) Show that, while B is moving in the positive x-direction, its motion satisfies

$$\frac{1}{v^2}\frac{\mathrm{d}v}{\mathrm{d}x} = -1 \qquad \textbf{[3]}$$

(b) (i) Solve the differential equation in part (a) to obtain v in terms of x and u. **[4]**

(ii) By considering the behaviour of v as $x \to \infty$, state one respect in which this model of the motion is unrealistic. **[1]**

The particle subsequently reaches a point A with coordinate X, where its speed is $V\,\text{m s}^{-1}$. The work done by the resistance as B moves from the origin to A is $W\,\text{J}$.

(c) (i) Using $W = \int F\,\mathrm{d}x$, where F is the signed force in the x-direction, find an expression for W in terms of X and u. **[3]**

(ii) Explain the physical significance of the sign of W. **[1]**

(iii) Rewrite your answer to part (c)(i) in terms of V and u, and hence show how W is related to the change in the kinetic energy of B. **[2]**

6. A small sphere P of mass $5\,\text{kg}$ is released into a liquid and moves vertically downwards. Take the downward direction as positive. In a model of the motion, the forces acting on P are:

- its weight;
- a constant upthrust of magnitude $14\,\text{N}$;
- a resistance to motion whose magnitude is proportional to the speed of P.

When the speed of P is $4\,\text{m s}^{-1}$, the resistance has magnitude $10\,\text{N}$. At time $t = 0$, the sphere is at a point O and is moving downwards with speed $u\,\text{m s}^{-1}$, where $u > 0$. At time t seconds, its downward velocity is $v\,\text{m s}^{-1}$. Take $g = 9.8\,\text{m s}^{-2}$.

(a) Show that, according to the model,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 7 - \frac{1}{2}v$$

[4]

(b) (i) Solve the differential equation in part (a) to show that

$$v = 14 + (u - 14)e^{-t/2}$$

[5]

(ii) Describe briefly how the speed of P varies over time in each of the following cases:

- $0 < u < 14$;
- $u = 14$;
- $u > 14$.

In each case, state the limiting speed of P as $t \to \infty$. **[3]**

(c) Given that $u = 9$, find the exact downward displacement of P from O when $t = 6$. **[4]**

7. A vehicle of mass 900 kg moves along a straight horizontal road. Its engine provides a constant driving force of 576 N. When the vehicle has speed $v\,\mathrm{m\,s^{-1}}$, the resistance to its motion has magnitude (v^2+14v) N.

The time taken for the speed of the vehicle to increase from $3\,\mathrm{m\,s^{-1}}$ to $13\,\mathrm{m\,s^{-1}}$ is T seconds.

(a) Show that

$$T = 18\ln\left(\frac{27}{7}\right)$$ **[6]**

When $t = 0$, the speed of the vehicle is $8\,\mathrm{m\,s^{-1}}$.

(b) Show that

$$\frac{v+32}{18-v} = pe^{qt}$$

where the rational numbers p and q are to be found. **[2]**

(c) Hence explain why $v < 18$ for all $t \geq 0$. **[1]**

8. A particle P of mass $2\,\text{kg}$ moves along the positive x-axis in the direction of increasing x. At time t seconds, its displacement from the origin is x metres and its speed is $v\,\text{m}\,\text{s}^{-1}$.

While P is moving, the resultant force acting on it is directed towards the origin and has magnitude $kv(v+2)\,\text{N}$, where k is a positive constant and v denotes the numerical value of the speed in metres per second.

When $x = 1$, the speed of P is $7\,\text{m}\,\text{s}^{-1}$, and when $x = 3$, its speed is $1\,\text{m}\,\text{s}^{-1}$.

(a) Find k and show that

$$v = 9 \cdot 3^{(1-x)/2} - 2$$ **[6]**

The time taken for the speed of P to decrease from $7\,\text{m}\,\text{s}^{-1}$ to $1\,\text{m}\,\text{s}^{-1}$ is T seconds.

(b) Show that

$$T = \frac{\ln(7/3)}{\ln 3}$$ **[5]**

9. A particle P of mass $4\,\text{kg}$ is released from rest and moves vertically downwards through a fluid. While P is moving with speed $v\,\text{m s}^{-1}$, it experiences a resistance of magnitude $6v\,\text{N}$ acting vertically upwards. After P has descended a distance $x\,\text{m}$, its speed is $v\,\text{m s}^{-1}$. Take $g = 9.8\,\text{m s}^{-2}$.

(a) Show that

$$\left(\frac{10v}{98 - 15v}\right)\frac{\mathrm{d}v}{\mathrm{d}x} = 1 \qquad \textbf{[2]}$$

(b) Verify that x and v satisfy

$$45x = 196\ln\left(\frac{98}{98 - 15v}\right) - 30v \qquad \textbf{[5]}$$

(c) Determine the work done against the resistance from the instant P is released until its acceleration has magnitude $2.45\,\text{m s}^{-2}$. Give your answer to 3 significant figures. **[5]**

10. A particle P, of mass m, moves on a rough straight track inclined at an angle α to the horizontal. A fixed point O lies on the track, and x denotes the distance of P from O, measured up the track.

In addition to its weight, P is acted on by a force directed down the track and having magnitude

$$\frac{kmg}{x+a}$$

where a and k are positive constants with dimensions of length. The coefficient of friction between P and the track is μ.

The particle is projected up the track and first comes to rest at the point where $x = 2a$.

(a) Show that, while P is moving up the track with speed v,

$$\frac{\mathrm{d}}{\mathrm{d}x}(v^2) + \frac{2gk}{x+a} + 2g(\sin\alpha + \mu\cos\alpha) = 0 \qquad \textbf{[4]}$$

(b) Verify that

$$v^2 = 2gk\ln\left(\frac{3a}{x+a}\right) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x) \qquad \textbf{[4]}$$

(c) Find, in terms of a, k and α, the range of values of μ for which P remains at rest when $x = 2a$. **[2]**

11. A small instrument package of mass $m\,\text{kg}$ enters a deep liquid moving vertically downwards at $7\,\text{m}\,\text{s}^{-1}$. While the package is moving downwards, it experiences a resistive force of magnitude $\frac{m}{20}v^2\,\text{N}$, where $v\,\text{m}\,\text{s}^{-1}$ is its speed.

At time $t\,\text{s}$ after entering the liquid, the package has descended a distance $x\,\text{m}$.

(a) Find an expression for v in terms of t. **[5]**

(b) Find an expression for x in terms of t. **[4]**

(c) Find the exact distance descended by the package when its speed first reaches $\frac{21}{2}\,\text{m}\,\text{s}^{-1}$. **[4]**

12. A package of mass m kg is released from rest and falls vertically. When its downward speed is $v\,\mathrm{m\,s^{-1}}$, it experiences a resistive force of magnitude mkv^2 N, where k is a positive constant. Take the downward direction as positive.

(a) Show that, at time t s after release,

$$v = \sqrt{\frac{g}{k}}\left(\frac{e^{2\sqrt{gk}\,t} - 1}{e^{2\sqrt{gk}\,t} + 1}\right) \qquad \textbf{[5]}$$

It is given that $k = 0.0125\,\mathrm{m^{-1}}$.

(b) Find the distance fallen by the package during the first 2 seconds after its release. **[4]**

13. A particle P of mass 16 kg moves in a straight line on a smooth horizontal plane.

At time $t = 0$, the particle passes through a fixed point O with velocity $\frac{5\sqrt{3}}{4}\,\mathrm{m\,s^{-1}}$.

From this instant, a horizontal force opposing the motion acts on P. The magnitude of the force is proportional to $\sqrt{25 - 4v^2}$, where $v\,\mathrm{m\,s^{-1}}$ is the velocity of P.

When $v = 2$, the magnitude of the force is 6 N. The force ceases to act when the velocity has decreased to $\frac{5}{4}\,\mathrm{m\,s^{-1}}$.

(a) Show that, while the force is acting, the motion of P is modelled by the differential equation

$$\frac{1}{\sqrt{25 - 4v^2}}\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{8}$$

[3]

(b) Hence find v in terms of t, showing that your expression is valid for

$$0 \leq t \leq \frac{2\pi}{3}$$

[5]

(c) Determine the total distance travelled by P during the interval $0 \leq t \leq 8$. Give your answer in an exact form. **[6]**

14. A vehicle of mass m kg moves along a horizontal straight road. Its engine works at a constant rate of $2kmg$ W, where k is a positive constant with units of speed. When the vehicle has speed $v\,\mathrm{m\,s^{-1}}$, it experiences a resistance of magnitude $\dfrac{mgv}{k}$ N.

Initially, the vehicle is at rest at $x = 0$, where x m is the distance travelled.

(a) Show that

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{k}\left(2k^2 - v^2\right) \qquad \textbf{[3]}$$

(b) Verify that

$$gx = -kv + \frac{k^2}{\sqrt{2}}\ln\left(\frac{\sqrt{2}k + v}{\sqrt{2}k - v}\right) \qquad \textbf{[5]}$$

(c) By using the work–energy principle, show that the time taken for the vehicle to accelerate from rest to a speed V, where $0 < V < \sqrt{2}k$, is

$$\frac{k}{2g}\ln\left(\frac{2k^2}{2k^2 - V^2}\right) \qquad \textbf{[8]}$$